

Customer Segmentation and Marketing Campaign Prediction

Project Background

A retail company has collected demographic, purchasing, and marketing campaign data from its customers.

The marketing department wants to better understand customer behavior and improve the effectiveness of future marketing campaigns.

As a Junior Data Analyst, your task is to analyze the data and build machine learning models that help the company make data-driven decisions.

Business Objectives

The company would like answers to the following questions:

Objective 1 – Customer Segmentation

Can customers be grouped into meaningful categories based on their characteristics and purchasing behavior?

Examples:

- High-value customers
- Frequent buyers
- Occasional customers
- Customers at risk of becoming inactive

This objective must be solved using **Unsupervised Learning**.

Objective 2 – Marketing Campaign Prediction

Can we predict whether a customer will accept a future marketing campaign? This objective must be solved using **Supervised Learning (Classification)**.

Dataset

You will use the Customer Personality Analysis dataset

The dataset contains information about:

- Customer demographics
- Income
- Family status
- Product spending
- Purchase behavior
- Previous marketing campaign responses

[Dataset link](#)

Project Requirements

Part 1: Data Cleaning and Preparation

Perform the following tasks:

1. Data Inspection

- Display the first rows of the dataset.
- Check data types.
- Check dataset dimensions.

2. Missing Values

- Identify missing values.
- Handle missing values appropriately.
- Explain your chosen method.

3. Data Type Conversion (If needed)

4. Feature Engineering

Create new variables that make sense and could help in our case study

Examples:

- $\text{Age} = \text{Current Year} - \text{Year_Birth}$
- $\text{TotalChildren} = \text{Kidhome} + \text{Teenhome}$
- $\text{TotalSpending} = \text{MntWines} + \text{MntFruits} + \text{MntMeatProducts} + \text{MntFishProducts} + \text{MntSweetProducts} + \text{MntGoldProds}$
- $\text{TotalPurchases} = \text{NumWebPurchases} + \text{NumCatalogPurchases} + \text{NumStorePurchases}$

Part 2: Exploratory Data Analysis

Perform exploratory analysis and answer the following questions.

Customer Demographics

1. What is the average customer age?
2. What is the most common education level?
3. What is the most common marital status?

Income Analysis

4. What is the average income?
5. How is income distributed among customers?
6. Is there a relationship between income and spending?

Spending Analysis

7. Which product category generates the highest spending?
8. Which product category generates the lowest spending?

Marketing Analysis

9. What percentage of customers accepted the latest campaign?

Visualizations

Create at least six visualizations, including:

- Histogram
- Bar Chart
- Pie Chart
- Boxplot

- Correlation Heatmap
- One additional visualization of your choice

For each visualization, provide a short interpretation.

Part 3: Customer Segmentation (Unsupervised Learning)

Objective

Identify different customer groups using clustering techniques.

Features

Use the provided features as a **baseline**. You may add additional features if you justify why they improve customer segmentation.

- Income
- Recency
- TotalSpending
- TotalPurchases
- NumWebVisitsMonth

Steps

1. Scale the selected features.
2. Determine the optimal number of clusters using the Elbow Method.
3. Apply K-Means clustering.
4. Assign a cluster label to each customer.

Analysis

Describe each cluster and provide a business interpretation.

Part 4: Marketing Campaign Prediction (Supervised Learning)

Objective

Predict whether a customer will accept the latest marketing campaign.

Target Variable :Response

Where:

- 1 = Customer accepted the campaign
- 0 = Customer did not accept the campaign

Data Preparation

1. Select relevant features.
2. Encode categorical variables.
3. Split the data into training and testing sets.
4. Apply scaling if necessary.

Models

Train and evaluate the following models:

Logistic Regression ,Decision Tree Classifier ,Random Forest Classifier

Evaluation Metrics

For each model report:

- Accuracy
- Precision
- Recall
- F1 Score

Comparison

Create a comparison table containing all evaluation metrics.

Identify the best-performing model and explain why it was selected.

Part 5: Business Insights and Recommendations

Assume you are presenting your findings to company management.

Answer the following questions:

Customer Segmentation

- Which segment appears to be the most valuable?
- Which segment requires additional attention?

Marketing Campaigns

- Which customers are more likely to accept campaigns?
- What characteristics do these customers share?

Recommendations

Provide at least three actionable recommendations supported by your analysis. Examples:

- Focus premium offers on high-spending customers.
- Send personalized campaigns to customers with high website activity.
- Re-engage inactive customers using targeted discounts.

Final Deliverables

Submit the following:

1. Jupyter Notebook

The notebook must include:

- Data Cleaning
- Feature Engineering
- Exploratory Data Analysis
- Customer Segmentation
- Marketing Campaign Prediction
- Conclusions

2. Final Report (PDF)

Maximum 8–10 pages.

The report must contain:

- Introduction
- Methodology
- Results
- Business Insights
- Recommendations

3. Presentation

Prepare a presentation of 8–12 slides summarizing your work.

Bonus Task

Choose one of the following:

Option 1

Train an additional classification model:

- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)

Compare its performance with the previous

models. **Option 2**

Create an interactive dashboard showing:

- Customer segments
- Spending analysis
- Campaign prediction results